

ANIMESH TIWARI

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Education

National Institute of Technology, Raipur

2020 - 2024

B.Tech in Information Technology

8.23 CGPA

Work Experience

ZeTheta Algorithms Private Limited

July 2026 - September 2026

Software Development Engineer Intern - Bangalore (Remote)

- Engineered a distributed payment orchestration system with intelligent multi-gateway routing and **2s automated failover**, achieving **99.95% transaction success rate**; implemented strict Distributed Idempotency using Redis TTL-based caching and transactional webhook reconciliation to prevent duplicate payouts.
- Designed an event-driven notification engine using Apache Kafka & RabbitMQ processing **50K+ daily financial events across 5 channels**; implemented exponential backoff retries and Dead Letter Queues (DLQ) to **reduce message drop rate to <0.01%**.
- Tech stack used:** TypeScript, JavaScript, Node.js, Express.js, React, PostgreSQL, Redis, Apache Kafka, RabbitMQ, Autocannon, K6 Testing, Docker, Nginx, Webhooks, WebSockets, Prometheus, Grafana, JWT, CI/CD, Github.

NR Engineering Works Pvt. Ltd.

November 2025 – May 2026

Software Development Engineer Intern - Hybrid

Certificate

- Built a full-stack workforce compliance management platform for a **300+ worker** manpower firm, architecting a **13-table** PostgreSQL schema with a rules-based payroll engine automating PF/ESI statutory deductions, **validated with 12 unit tests**.
- Designed an **ELT pipeline** using scheduled PostgreSQL materialized-view refreshes and Redis caching to power real-time analytics dashboards with **sub-3s** load times across workforce, cost, and payment data — replacing a fully manual, paper-based process.
- Tech stack used:** React, TypeScript, NestJS, Prisma, Redis, BullMQ, JWT, RBAC, Cloudflare R2, Docker, Jest & Github.

Research Internship

June 2023 – November 2023

Under Dr. Rakesh Tripathi (Dept. of IT, NIT Raipur)

IEEE Certificate Github

- Conducted research on lightweight deep learning models optimized for resource-efficient drowsiness detection systems.
- Optimized a lightweight MobileNetV2 architecture for edge deployment, achieving **94.03%** classification accuracy while compressing model footprint by **65% (down to 8.9MB)** to slash inference latency under low-resource constraints.
- Tech stack used:** Python (TensorFlow for Deep Learning Model), CNN, Multi-Model Testing and Git/GitHub.

Google Summer of Code

June 2022 – September 2022

Interned as an Open-Source Contributor - Remote

Certificate Github

- Developed Beacon, a Flutter-based cross-platform group travel management application, achieving a **100% decoupling** of business logic from UI components by designing a multi-layered Clean Architecture framework.
- Implemented MVVM architecture with GraphQL subscriptions for real-time location updates.
- Tech stack used:** Flutter, Dart, Node.js, GraphQL, GraphQL Subscriptions, WebSockets, CI/CD, Load Testing & Git/GitHub.

Projects

BRAHMO – Full-Stack Clinical AI System

Live Demo Github

- Built a full-stack clinical AI platform, engineering structured prompts and deterministic validation guards over Groq LLM outputs backed by a hybrid semantic cache (Redis), ensuring **100% compliance** with Indian clinical safety workflows.
- Designed responsive, doctor-facing dashboards using Next.js code-splitting and SSR, slashing First Contentful Paint (FCP) by **40% (down to 1.1s)** for seamless clinical workflow pilots.
- Tech stack used:** Next.js, TypeScript, PostgreSQL, Supabase, Groq API, TailwindCSS, CI/CD, Unit Testing, GIT/GitHub.

Fashion Intel – Fashion Brand News Monitoring Agent

Live Demo Github

- Architected a full-stack AI brand-intelligence agent, deploying a Node.js backend with 25+ APIs and an optimized vanilla JS live ticker that maintained a stable **60 FPS rendering cycle**.
- Eliminated **100%** of LLM pipeline parse failures by building a self-healing Zod validation layer over Groq's JSON-mode API, and restored accurate dataset attribution by redesigning the pipeline's deduplication logic.
- Tech stack used:** Node.js, Express.js, Vanilla JS, PostgreSQL, REST APIs, AI/LLM Integration, Zod, Unit Testing, Git/GitHub.

Technical Skills

Languages: Java, JavaScript, Python, TypeScript **Frontend:** HTML, CSS, React.js, Tailwind CSS, Bootstrap, Flutter
Backend: Node.js, Express.js, Next.js, REST APIs, FastAPI, Microservices, Webhook Handling, API Integration, Testing
AI Tools: LangChain, LangGraph, MCP, Retrieval-Augmented Generation (RAG), LLM, CrewAI, n8n, Prompt Engineering.
Databases, Cloud & Tools: MongoDB, Redis, SQL, PostgreSQL, GCP, AWS, Docker, GitHub, Postman, Apache Kafka
Computer Science Fundamentals: Data Structures & Algorithms, OOPS, Operating System, DBMS, AI, System Design

Positions of Responsibility

President - Royal Rook Chess Club, NIT Raipur

April 2022 – May 2024

- Served as the **President** of the Chess Club at NIT Raipur and was responsible for all chess related events.

President - ELITE Association of IT Department, NIT Raipur

Nov 2023 – May 2024

- Spearheaded all technical and cultural events for the IT department as Association President, optimizing student resources.

Achievements

- Selected in **Google Summer of Code 2022** in **CCEXtractor Development Organization**.
- Secured **13th Rank** in Codeforces DIV 2 Contest out of 17000 participants & Achieved a Rating of 2133 with a Master Title.
- Secured **1st Rank** in CodeChef DIV 1 (100 participants) and DIV 2 (1000 participants) contests respectively.
- Completed **Hacktoberfest 2021** Successfully with multiple pull requests from Organizations like **Appwrite, CCEXtractor, etc.**
- Participated in **Google Cloud Facilitator Program 2022**, and successfully completed it.